RESIDENTIAL BUILDING DESIGN

10-Marla House — Architectural Layout & 3D Massing Study

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1. Project Overview

This report documents the architectural layout and 3D massing study of a single-storey residential house, designed on a standard 10-Marla plot (35' x 70'). The project translates a dimensioned floor plan into a CAD-based 3D model to verify spatial proportions, circulation, fenestration (window/door placement), and the staircase geometry before the design is carried forward into structural detailing.

The model is composed of a clearly zoned plan — two drawing rooms, an open-plan kitchen and living area, twin washrooms, a long internal lobby, a front porch, and a dog-legged staircase — all developed on a single dimensioned grid and visualized through isometric massing views.

2. Site & Plot Information

Plot Size	35'-0" x 70'-0"
Plot Area	≈ 2,450 sq ft (10 Marla)
Building Footprint	Spans the full plot width, set back at the front porch
Floors Modeled	Single storey (ground floor), with parapet/roof line
Front Feature	Covered porch, 6'-10" x 9'-2"

3. Room Schedule

Dimensions were taken directly from the dimensioned floor plan and carried into the 3D model:

Space	Dimensions	Approx. Area
Drawing Room (x2)	12'-0" x 16'-0"	192 sq ft each
Washroom (x2)	7'-7" x 8'-0"	~61 sq ft each

Lobby / Corridor	6'-10" x 34'-6"	~236 sq ft
Porch	6'-10" x 9'-2"	~63 sq ft
Open Kitchen & Living Room	Combined open-plan area	Remaining floor area
Staircase	Tread 12" / Riser 6"	Per single-flight run

The plan is organized along a long internal lobby (6'-10" wide, running 34'-6") that threads together both drawing rooms, the washrooms, and the open kitchen/living zone — a layout that keeps circulation compact while giving each room independent access.

4. Staircase Design

The staircase is detailed with a 12-inch tread and a 6-inch riser. This is a comfortable, code-compliant proportion — the shallow 6" riser keeps the flight gentle and easy to climb, which is a deliberate choice for a residential, family-use staircase rather than a steeper utility stair.

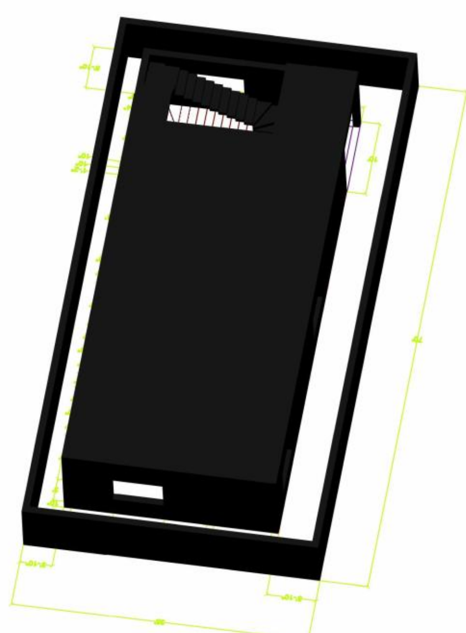

Fig. 1 — Isometric detail of the staircase, zoomed from the full 3D model.

5. 3D Massing Views

The full massing model was generated to check the building's overall form, opening placement, and roofline against the 2D plan before proceeding to structural drawings.

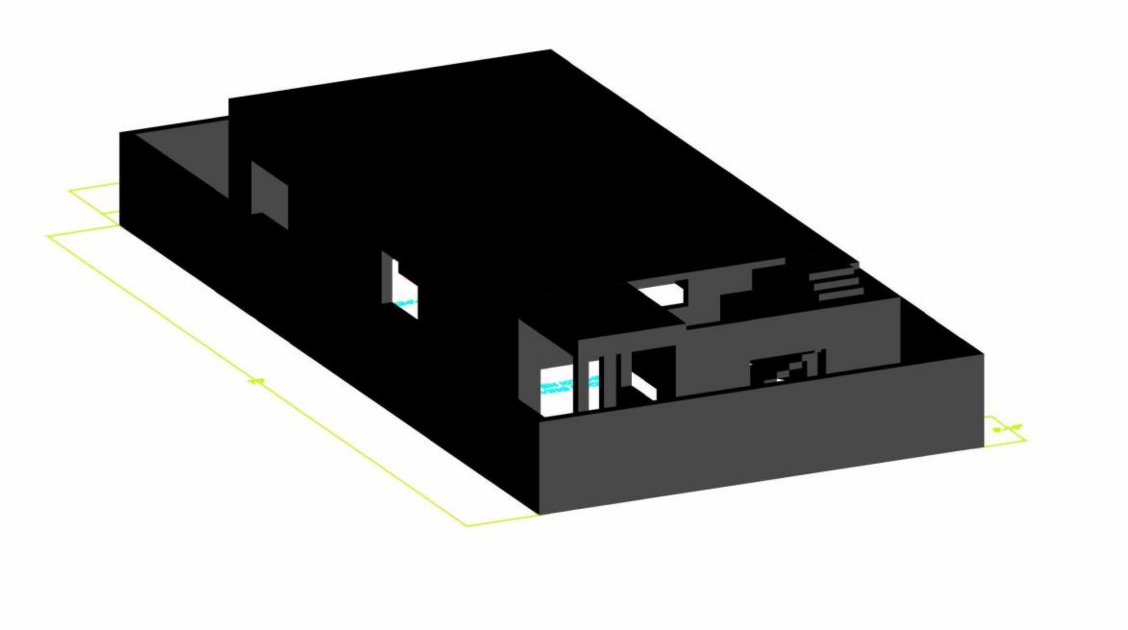

Fig. 2 — Front-facing isometric view of the building massing, showing window and door openings.

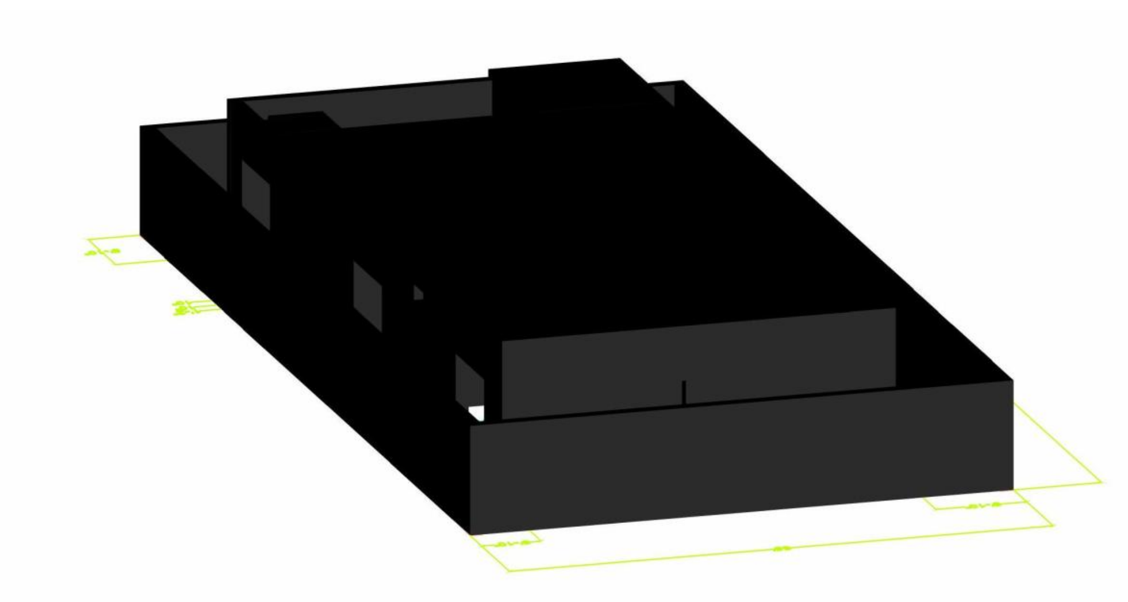

Fig. 3 — Rear isometric view showing the parapet line and porch volume.

6. Design Notes

- Twin drawing rooms (12' x 16' each) give the house two independent formal sitting spaces — a common requirement in local residential planning for separate family and guest use.
- The open kitchen and living room is left as one combined volume, maximizing usable open-plan space at the rear of the plan.

- A 6'-10" wide lobby was held consistently along its full 34'-6" run to keep circulation simple and column-free.
- The porch projects forward of the main building line, with a window cut into the side wall for natural light into the front room.
- The model was built directly over the dimensioned plan grid so that wall openings, the staircase, and the porch volume all align exactly with the working drawings.

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